

This was exceptionally close to human level performance. Even the human performer needed training to beat GoogleNet's accuracy which is indeed an accomplishment. This network utilised a CNN that was inspired by LeNet, however, it included a novel element that was dubbed an 'inception module'. GoogleNet used batch normalisation, RMSprop and image distortions. This architecture consisted of 22 deep layer CNN but reduced the number

of parameters from 60 million to a measly 4 million.

VGGNet

VGGNet was crowned the runner-up of ILSVRC 2014. It consists of a total of 16 convolutional layers and is appealing because of its consistent architecture. It

Year	CNN	Developed by	Place	Top-5 error rate	No. of parameters
1998	LeNet(3)	Yann LeCun et al			60 thousand
2012	AlexNet(7)	Alex Krizhevsky, Geoffrey Hinton, Ilya Sutskever	1st	15.3%	60 million
2013	ZFNet(3)	Matthew Zeiler and Rob Fergus	1st	14.8%	
2014	GoogLeNet(19)	Google	1st	6.67%	4 million
2014	VGG Net(16)	Simonyan, Zisserman	2nd	7.3%	138 million
2015	ResNet(152)	Kaiming He	1st	3.5%	

CNN summary and comparison table.

has 3x3 convolutions, much like AlexNet, but a lot more filters. VGGNet is currently the top choice in the community for extracting features from images. Additionally, the weight configuration of VGGNet is available to the public and has found home in many other applications and challenges as their baseline feature extractor. However, it consists of 138 million parameters which can prove to be challenging to handle.

ResNet

ResNet (2015) has beaten human level performance (on this dataset) with a top-5 error rate of 3.57%. It was unveiled at ILSVRC 2015 and introduced a unique architecture with 'skip connections' or gated units and features such as heavy batch normalisation. These gated units display a strong similarity to recent successful elements that were applied in RNNs. Due to this advanced technique, ResNet was able to train a Neural Network with 152 layers but still boasting lower complexity than VGGNet.

Future of Deep Learning

In the future we see certain trends and practices being adopted that will alter the deep learning technology and make it more efficient. Deep learning